Aggregation of Dialogue History Information and Utterance-Slot Attention for Dialogue State Tracking

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Abstract

Multi-domain Dialogue State Tracking (DST) is a critical component of task-oriented dialogue systems, designed to continuously update the dialogue state by integrating information from the current turn's utterance with the dialogue state from the previous turn. Traditional methods treat the previous turn's dialogue state as a holistic representation, frequently overlooking critical details from earlier turns and resulting in 'forgetting' issues. To address this limitation, we propose employing a discounted return algorithm to more effectively aggregate dialogue history information. Additionally, to capture slot-specific information within utterances, we introduce an utterance-slot attention mechanism that can model the semantics of the dialogue context while incorporating slot name features to guide the generation or extraction of the dialogue state. Our method achieved slot accuracy of 97.06% on the MultiWOZ2.1 dataset and 98.52% on the MultiWOZ2.4 dataset. The experimental results not only validate the superiority of our proposed method but also significantly enhance the accuracy and robustness of multi-domain dialogue state tracking.

1 Introduction

Task-oriented dialogue plays a crucial role in dialogue systems, enabling users to complete various tasks such as booking tickets, recommending music, and more. Dialogue State Tracking (DST) is a crucial component of task-oriented dialogue systems, aiming to extract key features from the dialogue history to determine the current state of the conversation accurately. Effective DST methods can enhance a system's ability to understand user intentions and provide more precise responses, thereby increasing user satisfaction and overall system performance. In DST, the dialogue state is viewed as a set of predefined (domain, slot, value) triples, such as the triple (restaurant-name, sitar tandoori), which indicates that the name of the restaurant is Sitar Tandoori, We consider the domain and slot names connected by a "-" as constraints on the slot. When the number of domains and slots remains constant, this setup allows the DST task to be viewed as a classification task.

In real-world scenarios, dialogues often span multiple domains. As shown in Figure 1, a single dialogue scenario may cover various domains, including restaurants and trains. In such multi-domain (Hong et al., 2023) settings, accurately tracking the current dialogue state poses significant challenges. Although state-of-the-art DST methods perform well, they typically only consider the utterance of the current turn and the dialogue state from the preceding turn (Zhang et al., 2019). This approach can lead to a "forgetting" (As the dialogue progresses, the dialogue state in the current turn relies on key information from previous turns) problem, where the model may overlook crucial information from earlier turns, especially in dialogues that cover multiple topics or domains. As demonstrated in Figure 1, the value for the slot "restaurant reservation time" cannot be derived solely from the current turn's dialogue. Instead, it necessitates reliance on previous dialogues to accurately determine the current dialogue state. Furthermore, when handling dialogues that involve complex slot interactions, traditional methods often struggle to capture the dependencies and dynamic changes between slots. For instance, in interactions involving multiple domains including hotels and restaurants, slots like 'hotel-star' and 'restaurant-price', and so on, are typically interdependent, with a change in one slot influencing the value of another.

Recent studies on DST models have primarily used inputs from the dialogue of the current turn and the state from the previous turn (Ni et al., 2023). However, since dialogue inherently builds upon previous exchanges, early rounds often contain crucial contextual clues. Thus, integrating utterance information from earlier turns is vital for accurately

Figure 1: The dialogue content covers two domains: restaurants and trains. In the first round of dialogue, it can be determined that the value of the slot 'train-day' is 'Sunday', and the value of 'train-arrive by' is '18:00'. In the fifth round of dialogue, the slot 'restaurant-book time' depends on the '18:00' mentioned earlier in the conversation. However, if we only consider the state of the dialogue from the previous round, the value of 'train-arrive by' is affected by the information in the intermediate rounds of dialogue, which leads to a 'forgetting' issue. The intermediate turns have been ignored.

generating dialogue states and building a detailed dialogue history. This approach of aggregating information from previous dialogues ensures the system comprehends the conversation's background and intent, leading to more coherent and relevant responses.

To address the "forgetting" problem in dialogue state tracking, we have introduced an innovative information aggregation method that utilizes a strategy based on discounted rewards from reinforcement learning (Wiering and Van Otterlo, 2012). This approach assigns different weights to various dialogue turns, effectively balancing information retention and the risk of forgetting. By prioritizing recent interactions while still acknowledging valuable information from earlier turns, our method not only mitigates the forgetting problem but also enhances the continuous tracking of critical information throughout the dialogue. This strategy ensures that essential contextual details are maintained, leading to more accurate and robust dialogue state tracking.

Inspired by previous work, we enhance our dialogue state tracking model by integrating an utterance-slot attention mechanism, enabling it to identify and utilize slot-related information within the dialogue more precisely. This mechanism enables the model to focus on slot-specific information within the sentence, thereby guiding the generation of slot values and enhancing the accuracy of slot value predictions, especially in situations where the dialog content is rich or there are numerous slots.

To effectively focus on key information from previous turns, we introduce a unique token [TURN] before each dialogue turn, which, similar to BERT's [CLS] token, aggregates rich representations of the current turn's dialogue utterances. We utilize the pre-trained encoder from BERT, which processes inputs comprising the utterance information from the dialogue history and the previous turn's dialogue state. This approach acknowledges that information closer to the current turn exerts a greater influence on the dialogue state of that turn. Inspired by the discounted reward algorithm, we dynamically allocate weights to the dialogue utterances of each turn. Furthermore, in order to focus on slot-specific information within dialogue utterances, we employ an utterance-slot attention mechanism. In this setup, predefined slots are encoded as query vectors, which engage in cross-attention calculations with the encoded dialogue utterance information. This process allows for the targeted extraction of slot-specific information from the sentences, optimizing the accuracy and relevance of the retrieved data. Our contributions are summarized as follows:

1. Inspired by the discount reward algorithm commonly used in reinforcement learning (Wiering and Van Otterlo, 2012), we propose a novel method to aggregate historical dialogue information. This approach is specifically designed to tackle the "forgetting" problem, which is prevalent in scenarios involving long dialogue dependencies. By assigning strategically weighted importance to earlier dialogue turns, our model maintains a more balanced and contextually enriched dialogue state.

2. We have implemented an utterance-slot attention mechanism that efficiently captures slot-specific information within dialogues. This mechanism not only enhances the accuracy of slot value prediction based on the contextual flow of the dialogue but also elucidates the interrelationships among different slots. By focusing on relevant portions of the dialogue pertinent to each slot, our model ensures a comprehensive understanding and representation of user intents and contextual nuances.

3. Our experiments validate the effectiveness of the proposed information aggregation method and the utterance-slot attention mechanism. Performance evaluations on the MultiWOZ 2.1 dataset demonstrated impressive outcomes, with slot accuracy reaching 97.06% and joint goal accuracy achieving 53.14%. Further testing on the MultiWOZ 2.4 dataset also showed enhanced results, with slot accuracy climbing to 98.52% and joint goal accuracy reaching 61.26%.

2 Related Work

Traditional pipeline-based approaches typically segment task-oriented dialogue systems into four distinct modules (Wang et al., 2022): Natural Language Understanding (NLU), Dialogue State Tracking (DST), Policy Learning (PL), and Natural Language Generation (NLG). Recently, the task of DST has gained increased attention due to the availability of large-scale dialogue datasets, such as MultiWOZ 2.0-2.4[1] and SGD[2], for free use. With the development of deep learning, research on DST tasks is primarily divided into two approaches: 1) DST based on predefined ontologies (Chen et al., 2020b),(Zhang et al., 2019),(Ye et al., 2021), and 2) methods based on open vocabulary (Shin et al., 2022),(Li et al., 2023),(Qiu et al., 2023),(Mao et al., 2021).

2.1 Based on predefined Ontology

The ontology-based approach (Chen et al., 2020b),(Zhang et al., 2019),(Ye et al., 2021) views the DST task as a classification problem. This method requires predefining all possible (domain, slot, value) triplets in the dialogue. It involves comparing the semantic features of the current turn against all possible (slot, value) pairs to calculate similarity scores, and then selecting the most suitable slot-value pair as the current dialogue state. To capture the features of the dialogue context and predefined slots, SUMBT (Lee et al., 2019) employs an attention mechanism based on contextual semantic vectors. This mechanism learns the relationships between the domain-slot types that appear in the discourse and the predefined slot values, enabling it to predict slot value labels in a non-parametric form. SST (Chen et al., 2020b),(Zeng and Nie, 2020b) introduces a pattern-guided sparse dialogue state tracking model that employs graph attention networks to predict slot values by integrating information from both discourse and structured graphs. However, in real-world applications, it is impractical to define ontologies for every conceivable slot. For instance, a taxi's destination could be a hotel or a restaurant, rendering it impossible to list all potential options, thereby compromising scalability and generalizability.

2.2 Open-vocabulary-based methods

To overcome the limitations of predefined ontologies in DST, recent studies have increasingly concentrated on open-vocabulary approaches (Zeng and Nie, 2020a),(Li et al., 2023),(Qiu et al., 2023), defining DST as an extraction or generation task. (Gao et al., 2019)redefines DST as a reading comprehension task, posing the question: "What is the current state of the dialogue?". (Shin et al., 2022) defines DST as a dialogue summarization problem, employing generative models with a soft

[1]https://github.com/smartyfh/MultiWOZ2.4

[2]https://github.com/google-research-datasets/dstc8-schema-guided-dialogue

copy mechanism that selectively copies words from the dialogue history and vocabulary to use as slot values. (Ren et al., 2019),(Su et al., 2023) selectively copies words from the dialogue history and vocabulary to use as slot values. With the introduction of graph structures, (Feng et al., 2022),(Yang et al., 2023),(Zeng and Nie, 2020b),(Su et al., 2023),(Huang et al., 2024) employs graph attention networks to enhance the model's ability to manage interactions across multiple domains and complex slots. (Li et al., 2023),(Xie et al., 2022) Another innovative approach involves modifiable state prediction, allowing for the correction of slot value predictions after initial generation to reduce error propagation in subsequent dialogues. With the emergence of large pre-trained language models, (Lin et al., 2021) leverages the powerful generative capabilities of GPT-2 combined with graph attention networks to capture dependencies between different domains in cross-domain dialogues, thereby overcoming the limitations of causal reasoning in GPT-2.

(Qiu et al., 2023),(Mao et al., 2021),(Yang et al., 2019),(Shan et al., 2020),(Wang et al., 2020) enhances adaptability to complex scenarios by dynamically modeling the variable dialogue environments and slots through various adaptations of attention mechanisms. (Kim et al., 2019),(Guo et al., 2021) views the dialogue state tracking task as a two-stage process: first determining which slot values need updating, and then generating new values for the slots that require updates. This approach eliminates the need to update all slot values at every dialogue turn.

3 Model

With the advancement of pre-trained language models like BERT (Devlin, 2018) and GPT-2 (Ham et al., 2020), which have shown remarkable effectiveness in various downstream tasks, we have adopted BERT as an encoder. This approach allows us to derive semantic vector representations from the dialogue context, as well as slots and values. The complete architecture of this system is depicted in Figure 2.

3.1 Overall

Encoder input: We define the dialogue state at each turn as $B_t = \{(S_t^j, V_t^j) j \in J]$, where S_t^j denotes the combination of the domain and slot for the j-th slot in turn t (e.g., "hotel - area"), and v_t^j represents the value for the j-th slot in turn t. Here, J symbolizes the total number of slots.

For each turn, we concatenate the user utterance and system response to form $D_t = (U_t + S_t)$, where U_t is the user input and S_t is the system response at turn t. We introduce the special token [TURN] before each dialogue turn. Additionally, slot-value pairs from the previous turn that are not NULL are also included as input to the encoder (Kim et al., 2019). Specifically,

$$X_t = D_t \cdot + \cdot B_{t-1} + [\text{TURN}] + \cdot D_{t-1} + [\text{TURN}] + \cdot + D_1 \tag{1}$$

Building on previous research, which utilized slot self-attention, we implemented a slot-utterance attention mechanism. This approach is designed to capture slot-specific information within the utterance, thereby effectively guiding the generation of slot values.

Encoder Output: The encoder's output representations are designated a $\boldsymbol{h}_t^{cls} \in R^{|seqlen|}$ and $h_t^{Turn_j} \in R^{|d|}$ Here, h_t^{cls} consolidates the output representation of the entire input sequence, serving as a comprehensive summary for the dialogue at turn t. We view h_t^{cls} as a detailed representation of the dialogue information for turn j and pass it to a subsequent information aggregation module. Furthermore, we employ a cross-attention mechanism between $h_t^{Turn_t}$. and all slot vectors h^{Slot} to effectively capture slot-specific information within the current dialogue context.

3.2 Information Aggregation Mechanism

Inspired by the discount factor used in reinforcement learning (Wiering and Van Otterlo, 2012), we acknowledge that dialogues further away from the current turn have a diminishing impact on the current dialogue state. To address this, we employ a technique known as weight decay to aggregate dialogue utterance information from previous turns. We define a decay rate r, which assigns weights to the dialogue information of each turn, reflecting the diminishing relevance of older dialogue information relative to the current turn. This method quantifies the importance of each turn's information, effectively prioritizing the most recent interactions. As shown in below

$$H_t^{urr} = h_t^{Turn_t} + r h_t^{Turn_{t-1}} + \ldots + r^{t-1} h_t^{Turn_1} \tag{2}$$

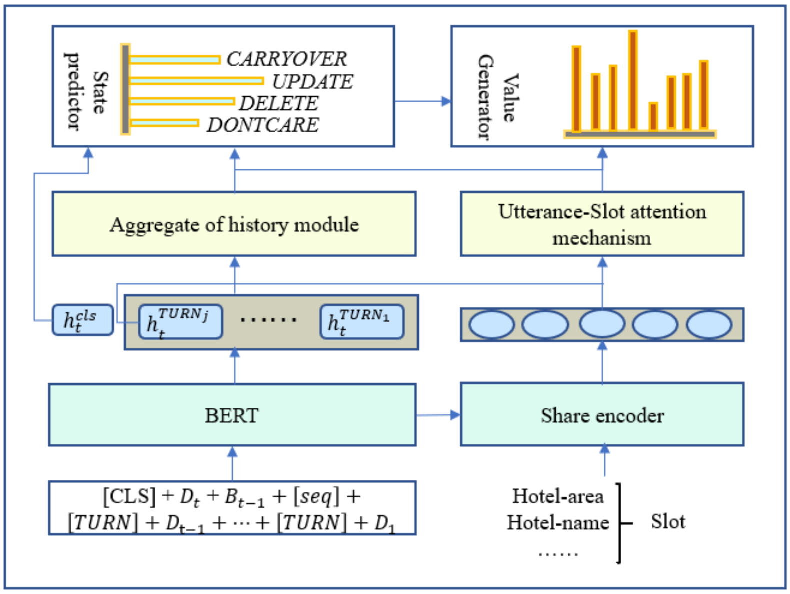


Figure 2: This describes the overall architecture of our dialogue state tracking model, which is built on an encoder-decoder framework. It includes an information aggregation module and a slot-utterance attention mechanism, utilizing a GRU as the decoder.

H_t^{urr} represents the aggregation of dialogue information from the current turn and previous turns, When the decay rate r approaches 0, we focus more on the utterances from the current turn of the dialogue. As r approaches 1, we place greater emphasis on integrating information from previous turns. In real dialogue scenarios, it is common to forget previous dialogue information as the conversation progresses. Our method of dialogue information aggregation, based on the concept of discounted rewards, mirrors the dynamic nature of human conversations. This approach helps to retain complex information from earlier dialogues, preventing its loss as the conversation continues.

3.3 Slot-Utterance Attention

To capture the correlation between the dialogue context and the corresponding slots, we propose the slot-utterance attention mechanism. By utilizing cross-attention, we can extract slot-specific information from the dialogue's contextual semantics. Based on the self-attention mechanism (Subakan et al., 2021), we use slot encoding as the query vector and apply self-attention with the utterance vectors to obtain the correlation between the dialogue's contextual semantics and the slots. We formalize this as follows:

$$H_t^{slot} = \mathrm{MultiHead}(Q_{slot}, k_{turn}, V_{turn}) \quad (3)$$

In this setup, Q represents the query vector derived from the slot encoding, while K and V are the key and value vectors extracted from the dialogue utterances. This mechanism enables us to effectively capture context-specific information pertinent to each slot, enhancing the model's ability to understand and respond to the nuances of the dialogue.

3.4 Value Generator

Following the setup from (Kim et al., 2019) we use a Gated Recurrent Unit (GRU) (Gu et al., 2020) as our decoder. We initialize $g_t^{j,0} = H_t^{urr}$ and $e_t^{j,0} = H_t^{slot_j}$. Here, $e_t^{j,k}$ represents the word embedding generated at each step, using the embedding generated in the current step $e_t^{j,k}$ as the hidden state until the end-of-sequence token [EOS] is generated.

$$g_t^{j,k} = \mathrm{GRU}\left(g_t^{j,k-1}, e_t^{j,k}\right) \quad (4)$$

The process can be summarized as follows:

(1) Initializing the GRU decoder with for the utterance context $g_t^{j,0} = \cdot H_t^{urr}$ and $e_t^{j,0} = \cdot H_t^{slot_j}$ for the slot information.

(2) At each decoding step k, generate the word embedding $e_t^{j,k}$.

(3) Use $e_t^{j,k}$ as the hidden state for the next decoding step.

(4) Continue the process until the end-of-sequence token [EOS] is generated, indicating the completion of the slot value generation.

The final distribution of the dialogue state is jointly influenced by two probabilistic components: the vocabulary probability distribution, denoted as $p^{j,k}_{vcb,t}$, and the dialogue text probability, denoted as $p^{j,k}_{ctx,t}$. The term $p^{j,k}_{vcb,t}$ specifically represents the hidden state $g^{j,k}_t$ at the k-th decoding step of the model, reflecting the influence of the selected vocabulary. On the other hand, $p^{j,k}_{ctx,t}$ is derived from the output that has been aggregated from the entire dialogue history up to the current turn, denoted by H^{urr}_t, following the information aggregation module's processing. This setup ensures that both immediate and historical dialogue contexts contribute to the generation of accurate and contextually appropriate dialogue states.

$$p^{j,k}_{vcb,t} = softmax\left(Eg^{j,k}_t\right)_{\downarrow} \tag{5}$$

$$p^{j,k}_{ctx,t} = softmax\left(H^{urr}_t g^{j,k}_t\right) \tag{6}$$

Here, E represents the word embedding for all the words in the dialogue. The final generated word is determined by summing the two probabilities mentioned above and selecting the word with the highest similarity.

$$p^{j,k}_{w,t} = \lambda p^{j,k}_{vcb,t} + (1-\lambda) p^{j,k}_{ctx,t^{\omega}} \tag{7}$$

3.5 Loss Function

In our experiments, our objective function is divided into two loss components: the state prediction phase and the slot-value generation phase. Our goal is to minimize the weighted sum of these two losses.

State Predictor: Here, we use the mean of the negative log-likelihood as the loss for state prediction, detailed as follows:

$$\mathrm{P}^{\mathrm{j}}_{\mathrm{op,t}} = softmax(\mathrm{W}_{op} H^{urr}_t) \tag{8}$$

$$\mathrm{L}_{\mathrm{op,t}} = -\frac{1}{\mathrm{J}} \sum_{\mathrm{j=1}} \mathrm{Y}^{\mathrm{j}}_{\mathrm{op,t}} \log \mathrm{P}^{\mathrm{j}}_{\mathrm{op,t}} \tag{9}$$

In this context, $P^j_{op,t}$ represents the prediction of the operation for slot j at the current turn, based on the aggregated message H^{curr}_t. This prediction is part of a four-category classification task. $Y^j_{op,t}$ denotes the true operation for slot j. J is the total number of predefined slots we are considering.

Value Generator: Similarly, in the slot value generation phase, we calculate the loss using the negative log-likelihood function, as follows:

$$\mathcal{L}_{gen,\mathrm{t}} = \sum_{j=1} \sum_{k=1} \mathrm{SVG}((\mathrm{Y}^{j,k}_{w,t}) \log(p^{j,k}))_{w,t} \tag{10}$$

In this case, k represents the number of tokens in the encoded slot value, and j denotes the slots predicted to be in the "update" state. $Y^{j,k}_{w,t}$ represents the true value of the current slot value, encoded as a one-hot vector.

Finally, our target loss is the sum of the state prediction loss and the slot value generation loss, which can be expressed as:

$$\mathrm{L}' = \mathrm{L}_{gen,\mathrm{t}} + \mathrm{L}_{op,\mathrm{t}} \tag{11}$$

4 Experiments

4.1 Datasets

We evaluate our model on the publicly available task-oriented dialogue datasets, which are MultiWOZ 2.1 and MultiWOZ 2.4. MultiWOZ 2.0, which is a task-oriented, human-to-human written dialogue dataset that spans seven domains, consists of 10,438 multi-turn dialogues, with an average of approximately 13 turns per dialogue. MultiWOZ 2.1, which is a revised version of the MultiWOZ 2.0 dataset, corrects some annotation errors in MultiWOZ 2.0. MultiWOZ 2.4, which represents an enhanced variant of MultiWOZ 2.1, includes fine-grained annotations for the validation and test sets, thereby improving the overall quality of the dataset. Because there are no dialogues in the validation and test sets for the hospital and police domains, we follow the framework established by TRADE and use dialogues from the hotel, taxi, train, attraction and restaurant domains to evaluate our model. These five domains include a total of 17 predefined slots, resulting in 30 (domain, slot) pairs, as detailed in Table 1.

4.2 Baseline

In the field of multi-domain dialogue state tracking, since obtaining a predefined ontology in real-world scenarios is challenging, these methods may not

Table 1: Information on the MultiWOZ 2.1 dataset is provided below. The predefined domains and slots in the MultiWOZ 2.4 dataset are the same as those listed.

Domain	Slot	Train	Val	Test
Restaurant	area, food, name, price range, book day, book people, book time	3813	438	437
Hotel	area, internet, name, parking, price range, stars, type, book day, book people, book stay	3381	416	394
Train	Arrive by, day, departure, destination, leave at, book people	3103	484	494
Taxi	arrive by, departure, destination, leave at	1654	207	195
Attraction	area, name, type	2717	401	395

apply to actual situations. Therefore, we evaluated the performance of our model by comparing it to models that are based on open vocabularies.

TRADE (**Wu et al., 2019**) utilizes a generative model that operates across multiple domains, reducing reliance on domain-specific data. It employs a bidirectional GRU to encode the entire dialogue and incorporates a soft copy mechanism during the decoding process to accurately retrieve slot values.

PIN (**Chen et al., 2020a**) employs an interactive encoder to capture both intra-turn and inter-turn information effectively. It introduces a slot-level context extraction method to differentiate features across various slots and utilizes a distributed copy mechanism to selectively copy words from the dialogue history.

Som-DST (**Kim et al., 2019**) proposes treating the dialogue state as fixed-size memory and introduces a selective updating mechanism that generates only partial slot values each turn. It employs a Bert encoder to process both the current turn's dialogue and the previous turn's dialogue state. It predicts predefined slot operations for each slot and then utilizes a GRU decoder to generate the slot values.

DST-STAR (**Ye et al., 2021**) introduces a slot self-attention mechanism that captures slot-specific features within the dialogue context. By learning correlations between slots, it effectively updates slot values through a slot value matching mechanism.

Seq2Seq (**Feng et al., 2020**) employs a sequence-to-sequence model to manage dialogue state tracking tasks.

RefPyDST (**King and Flanigan, 2023**) redefines dialogue state tracking as a Python programming challenge within a context learning framework, achieving excellent results in low-sample learning scenarios.

4.3 Training Details

We utilize the pre-trained bert-base-uncased model as our encoder, which serves as the backbone for the slot value generator, employing a bidirectional GRU. For the encoder, we set the dropout to 0.1, the warmup to 1e-4, and the learning rate to 1e-5. For the decoder, we set the learning rate to 1e-4. In our information aggregation mechanism, we conducted comparative experiments with decay rates of 0.9 and 0.1 to assess the impact of dialogue history information on slot value generation, with specific hyperparameters detailed in Table 2. We used the model that achieved the best results on the validation set for testing.

Table 2: Hyperparameters during the training process.

Hyperparameter	Explanation	Value
enc_lr	Encoder learning rate	1e-5
dec_lr	Decoder learning rate	1e-4
warmup	Warm-up parameters	0.1
dropout	Randomly dropout	0.1
r	Decay rate	0.1/0.5/0.9

5 Results

5.1 Evaluation Metrics

Slot Accuracy and Joint Goal Accuracy, which are used to evaluate task-oriented dialogue systems and are considered two important metrics, are the criteria that we employ for our evaluation.

Slot Accuracy(SA), which is used to evaluate the accuracy of a dialogue system in recognizing and extracting slots within a user's intent, where slots refer to specific pieces of information such as the date for a hotel reservation or the hotel name, represents the proportion of correctly recognized slots, and is calculated as follows:

$$SA = \frac{\sum_{t=0}^{n} \frac{K-M_t}{K}}{n} \tag{12}$$

Where K is the number of predefined slots, n is the total number of dialogue turns, and M_t is the number of incorrectly predicted slots in the current turn t.

Joint Goal Accuracy(JGA), which refers to whether a dialogue system can correctly identify and maintain all slots related to the user's intent throughout the dialogue, requires that every step in the dialogue process correctly recognizes and tracks the status of all slots, and is calculated as follows:

$$JGA = \frac{\sum_{t=0}^{n} (1 \mid G_t = P_t)}{n} \tag{13}$$

Where n is the total number of dialogue turns , $\mathbf{G}_t$ and $\mathbf{p}_t$ represent the real slot status and the predicted slot status respectively. It is recorded as 1 if and only if all slot statuses of the current turn t are accurately predicted.

In a single dialogue that includes multiple turns, if any turn fails to correctly recognize the status of all slots, then that turn cannot be included in the JGA.

5.2 Experimental Results

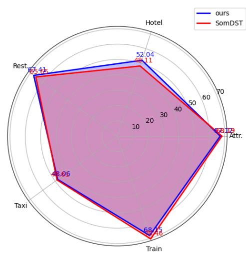
To evaluate the effectiveness of our model, we conducted experiments, which were validated using the MultiWOZ 2.1 and MultiWOZ 2.4 datasets, and the results of these experiments are shown in Table 3.

Based on the results in Table 3, we have reached the following two conclusions: 1) Compared to the baseline model, our model still needs improvement in Joint Goal Accuracy and has not yet achieved optimal performance. However, the Slot Accuracy of our model is close to or has already achieved optimal performance. We hypothesize that adding additional contextual information during the inference process introduces noise. This noise affects our model's ability to infer certain slot values based solely on the current dialogue round, resulting in slots that originally had NULL values being erroneously filled with new values. 2) Compared to the baseline model SOM-DST (Kim et al., 2019), our model incorporates an information aggregation module and a discourse-slot attention mechanism, significantly outperforming SOM-DST across both datasets. This improvement suggests that the integration of these two modules effectively enhances performance.

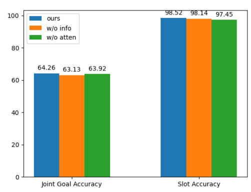
5.3 Experimental Analysis

To further validate the effectiveness of our proposed information aggregation module based on the discounted reward algorithm, we present an example in Tabel 4. Tabel 4 shows that our method can aggregate dialogue history information, which is beneficial for obtaining slot values that cannot be directly retrieved from the turn round of dialogue. The reason is that the dialogue history contains parts related to predefined slots, which can effectively supplement the previous dialogue state. However, our method also introduces some noise, as it fills slots with invalid values when the true value for the current round state is NULL, thereby reducing the accuracy of the joint goal. We may need to design a better information aggregation mechanism, which can address the "forgetting" problem in the DST domain while not introducing noise.

We present the joint goal accuracy of our method in specific domains in figure 3a, where the results indicate that the combination of the information aggregation module and the utterance-slot attention mechanism greatly benefits the hotel and restaurant domains. In the MultiWOZ 2.4 dataset, the hotel and restaurant domains always appear alongside another domain, making their slot values consistently related to the dialogue information from previous turns. Our method aggregates the historical information of the dialogue, effectively capturing information related to slot values across multiple rounds of dialogue with domain transitions, thereby addressing the slot value forgetting problem in multi-domain dialogue state tracking. However, our method performs poorly in the other three domains, such as the taxi domain, where slots like "leave-at" and "arrive-by" are heavily influenced by the current round of dialogue, and our information aggregation mechanism, which introduces additional information, generates noise when guiding slot value generation.



(a) Joint goal accuracy for each domain



(b) Ablation experiments

Figure 3: Figure 3a shows the joint goal accuracy for each domain on the MultiWOZ 2.4 dataset. Figure 3b shows the ablation experiments on the MultiWOZ 2.4 dataset, where w/o indicates 'without'.

Table 3: The results of our experiments were validated on the MultiWOZ 2.1 and MultiWOZ 2.4 datasets. * indicates the result of our reproduction, while the other results are taken from the original paper.

model	Joint Goal Accuracy(%)		Slot Accuracy(%)	
	MultiWOZ 2.1	MultiWOZ 2.4	MultiWOZ 2.1	MultiWOZ 2.4
DS-DST(Zhang et al., 2019)	51.21	-	96.62	-
TRADE(Wu et al., 2019)	48.62	55.05*	96.92	97.26*
PIN (Chen et al., 2020a)	48.40	55.92	97.02	98.02
SOM-DST(Kim et al., 2019)	53.68	63.21*	**98.38**	98.41*
DST-STAR(Ye et al., 2021)	**56.36**	-	95.53	-
Seq2Seq(Feng et al., 2020)	52.80*	**67.10***	-	-
RefPyDST(King and Flanigan, 2023)	52.0	65.2	95.84*	98.25*
OURS	53.14	64.26	97.06	**98.52**

Table 4: In my testing instances on the MultiWOZ 2.4 dataset, the slot values are marked with different colors to denote their inferential sources: green indicates that the slot values can be inferred from the current round of dialogue, blue signifies that the slot values are deduced from the dialogue state of the previous round, and red requires the combination of dialogue utterances from prior rounds for accurate inference.

Turn	Dialogue
1	**U:** I am looking for a train that leaves on Saturday and arrives by 10:30. **S:** Where are you traveling to and from? **State:** ["train-day-Saturday", "train-arrive by-10:30"]
2	**U:** I am going to Cambridge from Birmingham New Street. **S:** Here are 5 trains available; may I book 1 for you that leaves at 7:40 and arrives at 10:23? **State:**["train-destination-Cambridge","train-day-Saturday", "train-arrive by-10:30", "train-departure-Birmingham new street"]
3	**U:** yes , that train sounds good . please book it for me. could you also find me a hotel with a moderate price that offers internet ? **S:** there are 17 hotel -s listed. what part of town would you like to stay in? **State:** ["hotel-price range-moderate", "hotel-internet-yes", "train-destination-Cambridge", "train-day-Saturday", "train-arrive by-10:30", "train-departure-Birmingham new street"]
4	**U:** The north part of town please; preferably in a guesthouse. **S:** Cambridge has 8 moderate -ly priced guesthouses in the north. I would suggest acorn guest house, with a star rating of 4. Would you like me to book it? **State:** ["hotel-area-north","hotel-price range-moderate", "hotel-internet-yes","hotel-type-guest house","train-destination-Cambridge", "train-day-Saturday", "train-arrive by-10:30", "train-departure-Birmingham new street"]
5	**U:** yes please , i would like to book it for7 people for 5 nights, and i need a reference number **S:** your booking was successful, the reference number is waeyaq0m **State:** ["train-day-Saturday", "train-departure-Birmingham new street", "train-destination-Cambridge", "hotel-internet-yes", "hotel-price range-moderate", "hotel-area-north", "hotel-type-guest house", "hotel-book day-Saturday", "hotel-book people-7", "hotel-book stay-5", "hotel-name-acorn guest house"]

5.4 Ablation Experiments

Meanwhile, we present in Figure 3b the joint goal accuracy of the model on the MultiWOZ 2.4 dataset, comparing the results with and without the information aggregation module and the utterance-slot attention mechanism, relative to the base model Som-DST, to determine the impact of each module on model performance. The results indicate that our proposed information aggregation module and discourse-slot attention mechanism are beneficial to the dialogue state tracking model. To determine the impact of the information aggregation module on model performance, we use only the dialogue utterances from the current round and the dialogue state from the previous round as input. It can be seen that our joint goal accuracy, which is lower than that of our baseline model, indicates that our information aggregation module is beneficial to the overall performance of the model, even though it introduces some noise.

To investigate whether the utterance-slot attention mechanism can effectively guide slot value generation, we remove the utterance-slot module and initialize the slot value generator state as the encoder output h_t^{cls}. During the experiment, we found that without the utterance-slot attention mechanism, our slot accuracy decreased by 1.62 percentage points. Our utterance-slot attention mechanism can capture specific information in the sentences, effectively extracting the corresponding slot values from the dialogue.

5.5 Comparative experiments

To further explore the effectiveness of our proposed dialogue information aggregation method based on discounted returns, we conducted a series of comparative experiments with different reward decay rates, setting r at 0.1, 0.5, and 0.9 to assess how varying degrees of decay intensity affect model performance, with detailed experimental results presented as follows Table 5.

Table 5: We conducted experiments on the MultiWOZ 2.4 dataset by setting different decay rates to aggregate historical dialogue utterances

r	SA(%)	JGA(%)
0.1	97.22	**61.26**
0.5	96.43	60.01
0.9	**98.52**	58.47

Upon examining the results in Table 5, we observe that a high Joint Goal Accuracy is achieved as r approaches 0. Conversely, as r nears 1, there is an increase in Slot Accuracy. Our analysis suggests that, with an average of 8.93 dialogue turns in the dataset, the noise may be introduced for slots with NULL values when r is close to 1, leading to a decrease in Joint Goal Accuracy as r increases. However, our method effectively addresses the forgetting problem of slot values, which increases with the number of dialogue turns, thereby enabling the correct generation of slot values that might otherwise be forgotten.

6 Conclusion

In this paper, we propose an information aggregation method for dialogue state tracking tasks based on discounted rewards. We employ the concept of the discounted return algorithm to aggregate historical dialogue information. The utterance-slot attention mechanism effectively guides the generation of slot values. We conducted extensive experiments to validate the effectiveness of these two modules, exploring the joint goal accuracy across five domains to determine whether the information aggregation module is effective. The combination of results indicates that the method we proposed has a positive impact on the generation of slot values in guiding multi-turn, cross-domain dialogues. However, as the aggregation of historical dialogue utterances inevitably results in unwanted noise, our subsequent research will focus on how to minimize this noise.

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A Example Appendix

This is an appendix.